

Question 1: E-commerce Customer Dataset Cleaning

You are given a dataset customers.csv from an online shopping platform with the following columns:

- CustomerID (string)
- Age (numeric but stored as string, some missing)
- Gender (Male/Female, some missing)
- Annual_Income (contains values like "50,000", "not available")
- City (categorical, some nulls)
- Duplicate rows exist

Tasks:

1. Load the dataset using pandas
2. Detect missing values in all columns
3. Remove rows where more than 2 values are missing
4. Fill missing Age using median
5. Fill missing City using mode
6. Convert Annual_Income into numeric (handle errors properly)
7. Remove duplicate rows
8. Convert Gender into numerical form using appropriate encoding

Question 2: Hospital Patient Records (Data Types + Encoding)

A hospital provides a dataset patients.csv:

- Patient_ID
- Admission_Date (different formats: "01-02-2024", "2024/03/01")
- Discharge_Date
- Severity (Low, Medium, High)
- Department (Cardiology, Neurology, Orthopedic)
- Some invalid dates and missing values

Tasks:

1. Convert Admission_Date and Discharge_Date to datetime format
2. Handle invalid dates using appropriate method
3. Encode Severity using ordinal encoding
4. Encode Department using one-hot encoding
5. Check and fix any incorrect data types

Question 3: House Price Prediction (Feature Scaling)

You are working on a machine learning model to predict house prices. Dataset housing.csv contains:

- Area (in square feet, range: 500–5000)
- Bedrooms (1–5)
- Price (target variable)
- Distance_from_City (in km)
- Age_of_House

Tasks:

1. Identify which features need scaling
2. Apply **Normalization** on Area and Distance_from_City
3. Apply **Standardization** on Age_of_House
4. Train-test split the data
5. Apply scaling correctly (fit on train, transform on test)

Question 4: Food Delivery Dataset (Duplicates + Missing + Encoding)

Dataset orders.csv contains:

- Order_ID
- Restaurant_Name
- Cuisine_Type (Italian, Chinese, Pakistani)
- Delivery_Time (minutes, some missing)
- Customer_Rating (1–5, some missing)
- Duplicate orders exist

Tasks:

1. Detect and count duplicate rows
 2. Remove duplicate records
 3. Fill missing Delivery_Time using mean
 4. Fill missing Customer_Rating using median
 5. Encode Cuisine_Type using one-hot encoding
 6. Check final dataset consistency
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Question 5: Student Performance Dataset (Full Preprocessing Pipeline)

Dataset students.csv contains:

- Name
- Marks (some missing, some stored as text like "85 marks")
- Grade (A, B, C, D)
- Study_Hours
- Passed (Yes/No)
- Some duplicate rows

Tasks:

1. Clean Marks column and convert it to numeric
 2. Handle missing values in Marks using mean
 3. Remove duplicate rows
 4. Encode:
 - Grade using ordinal encoding
 - Passed using label encoding
 5. Apply standardization on Study_Hours
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Question 6: Online Course Platform (Mixed Data Issues)

Dataset:

```
import pandas as pd
```

```
data={
    'Student_ID': [1, 2, 3, 4, 2],
    'Course': ['AI', 'Web Dev', 'AI', None, 'Web Dev'],
    'Completion_Percentage': ['80', '90%', None, '70', '90%'],
    'Enroll_Date': ['2024-01-10', '10/02/2024', 'invalid', '2024-03-01',
                    '10/02/2024'],
    'Feedback': ['Good', 'Excellent', 'Average', 'Good', 'Excellent']
}
```

```
df=pd.DataFrame(data)
print(df)
```

Tasks:

1. Detect missing values in the dataset
2. Remove duplicate rows
3. Clean Completion_Percentage (remove % and convert to numeric)
4. Convert Enroll_Date into datetime and handle invalid values
5. Fill missing values:
Completion_Percentage → mean
Course → mode
6. Apply:
One-Hot Encoding on Course
Label Encoding on Feedback
7. Apply **Normalization** on Completion_Percentage

Question 7: Retail Store Sales Dataset (Full ML Preprocessing)

Dataset:

```
import pandas as pd
```

```
data={
'Product_ID': [101, 102, 103, 104, 101],
'Category': ['Electronics', 'Clothing', 'Electronics', 'Grocery',
'Electronics'],
'Price': ['1000', '2000', 'invalid', '500', '1000'],
'Quantity_Sold': [5, 10, None, 8, 5],
'Discount': [10, 20, 15, None, 10],
'Purchase_Date': ['2024/01/01', '01-02-2024', '2024-03-01', 'invalid',
'2024/01/01']
}
```

```
df=pd.DataFrame(data)
print(df)
```

Tasks:

1. Identify and remove duplicate rows
2. Convert Price into numeric and handle invalid values
3. Convert Purchase_Date into datetime
4. Handle missing values:
Quantity_Sold → median
Discount → mean
5. Apply:
One-Hot Encoding on Category
6. Apply **Standardization** on:
Price
Quantity_Sold

7. Perform train-test split
8. Apply scaling correctly (fit on train, transform on test)

Question 8: Banking Customer Risk Analysis

A bank wants to analyze customer data to predict loan risk. The dataset contains the following columns:

- Customer_ID
- Age (may contain missing values and incorrect entries like "unknown")
- Income (stored as string with commas, e.g., "50,000")
- Loan_Status (Approved, Rejected)
- Credit_Score (some missing values)
- City (categorical, some null values)
- Duplicate records may exist

Tasks:

1. Detect missing values in all columns
2. Remove duplicate records
3. Convert Income into numeric format
4. Clean and convert Age into numeric (handle invalid values)
5. Handle missing values:
 - Age → median
 - Credit_Score → mean
 - City → mode

6. Apply:

Label Encoding on Loan_Status

One-Hot Encoding on City

Apply **Standardization** on Income and Credit_Score
